

Comparison of Seattle and New York Precipitation

1. Introduction

This project compares daily precipitation patterns between Seattle, Washington and New York City (airport station) during 2018. The goal is to understand how precipitation differs in frequency, intensity, and seasonal variation between a Pacific Northwest maritime climate and an East Coast continental climate. Two daily weather datasets were obtained from the NOAA Global Historical Climatology Network (GHCN), containing fields such as STATION, DATE, and PRCP (precipitation). Data from all New York stations were filtered to retain only the airport station, ensuring consistent comparison with the single Seattle station.

2. Methods

Data cleaning and analysis were conducted in Python using the pandas library. The `info()` method was used to confirm column data types, identify missing values, and verify dataset size. Both datasets included approximately 365 daily records with numeric precipitation (PRCP) values. The DATE column was converted to a datetime object, and data were merged using an inner join on DATE. Filtering identified rainy days ($PRCP > 0$). Monthly summaries and four plots were created for visualization: (1) Daily precipitation series, (2) Histogram of Seattle precipitation, (3) Histogram of New York precipitation, and (4) Monthly rainy-day counts.

3. Analysis and Findings

Seattle exhibits frequent low-intensity rainfall, typical of a maritime climate. New York shows fewer but higher-intensity events. Seattle's rainfall is steady through winter, while New York's precipitation peaks in spring and summer. Seattle recorded about 170–180 rainy days compared to New York's 110–130.

4. Discussion

The contrast between the two cities highlights climate differences: Seattle's marine climate yields steady, light rain, while New York's continental climate produces fewer but heavier storms. Both datasets were adequate for analysis after cleaning and station filtering.

5. Conclusion

Seattle experiences more frequent but lighter rainfall throughout the year, while New York experiences fewer but heavier downpours. Seasonal differences show Seattle peaking in late fall and winter, and New York peaking in spring and summer. In summary, Seattle is wetter by frequency, New York by intensity.

6. References

NOAA Global Historical Climatology Network (GHCN):
<https://www.ncdc.noaa.gov/ghcn-daily-description>
Matplotlib Documentation: <https://matplotlib.org/stable/>
pandas Documentation: <https://pandas.pydata.org/docs/>